

Amr Wael

Flutter Developer | Mobile & Full-Stack Software Engineer

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PROFESSIONAL SUMMARY

Flutter Developer with production experience building and shipping cross-platform mobile applications using Flutter and Dart. Founder and sole developer of Za2zo2a, a full-stack bilingual ride-hailing platform (rider app, driver app, and Flutter Web admin panel) featuring real-time WebSocket trip tracking, an in-ride safety suite, and Paymob payment integration. Strong in Clean/feature-first architecture, BLoC/Cubit state management, RESTful API integration, and Firebase services, with hands-on Node.js/TypeScript/MongoDB backend delivery. Focused on clean, tested, maintainable code and user-centric mobile solutions.

TECHNICAL SKILLS

- Languages:** Dart, TypeScript, Java, C++, C#
- Mobile Development:** Flutter, Flutter Web, Cross-Platform Development, Responsive UI, Material & Cupertino Widgets, Bilingual i18n (Arabic/English), Light & Dark Theming
- State Management & Architecture:** BLoC/Cubit, Provider, GetX, Clean Architecture, Feature-First Architecture, MVVM, Repository Pattern, Dependency Injection (get_it), go_router, OOP
- Backend & Databases:** Node.js, Express, TypeScript, MongoDB/Mongoose, WebSocket, JWT Authentication, REST API Design, Firebase (Auth, Firestore, Storage, Cloud Messaging), Realtime Database, SQLite, Cloudinary
- Integrations:** Paymob Payment Gateway, OpenStreetMap/flutter_map, TomTom & Nominatim Geocoding, Geolocator/GPS, Push Notifications, Nodemailer/Brevo, OTP Verification Flows
- APIs & Networking:** RESTful APIs, dio, GetConnect, JSON Parsing, Interceptors, Error Handling, Rate Limiting, Helmet & CORS
- Testing & Tools:** Unit & Widget Testing, Git, GitHub, Postman, Agile/Scrum, easy_localization, Performance Optimization, AI-Assisted Development
- Game Development:** Unity, Blender 3D, C#, 2D & 3D Game Design

PROFESSIONAL EXPERIENCE

- AI Integration Intern • Tips Hindawy** Jan 2026 – Feb 2026
- Integrated AI features into mobile applications, supporting intelligent and adaptive user experiences.
 - Worked with data processing and model interaction to implement AI-based functionalities in real projects.
 - Collaborated with the development team to explore and apply AI solutions to real-world problems.
- Flutter Developer Intern • Ideas and Concepts** Jun 2025 – Aug 2025
- Built and shipped a production field-merchandising app (Merchandiser App) deployed for use in Saudi Arabia, supporting merchandisers across store visits, task management, and reporting.
 - Developed the app end-to-end using GetX for state management and routing, with a REST API layer integrating auth, markets, branches, tasks, products, RTV, and planogram modules.
 - Implemented an attendance system, before/after ordering photo tasks, and inventory/availability checks, with full Arabic/English localization (easy_localization), light/dark themes, and a responsive UI (flutter_screenutil).
 - Worked within a collaborative team, applying clean, scalable code practices throughout.
- Gaming Developer Intern • Thaka International** Jul 2023 – Sep 2023
- Designed and developed 2D & 3D games using Unity and C#, covering core game mechanics, physics, and animations.
 - Built a Maze Game in Unity with C# game logic and 3D assets modeled in Blender, and optimized game UI/UX for smooth performance across target platforms.

PROJECTS

- Health Rootz — Remote Health-Monitoring App (Graduation Project)**
Flutter, Dart, flutter_bloc, dio, Firebase, REST API, IoT Sensors
- Built the complete Flutter app with a multi-tenant architecture serving two separate experiences (Patient & Doctor) in a single codebase, with fully isolated state across 20+ Cubits.
 - Engineered a centralized dio networking layer with interceptors for automatic JWT injection, a repository pattern for unified error handling, and global 401 logout/redirect handling.
 - Implemented hardware-integration UX with countdown and polling-with-backoff for live sensor readings, interactive fl_chart visualizations, a responsive layout (mobile/tablet/desktop), and full Arabic/English localization.
- Za2zo2a — Full-Stack Ride-Hailing Platform (Founder & Sole Developer)**
Flutter, Dart, flutter_bloc (Cubit), go_router, get_it, dio, flutter_map/OpenStreetMap, geolocator, web_socket_channel, Node.js, Express, TypeScript, MongoDB, Paymob, Cloudinary, Firebase
- Founded and conceived the product to solve Egypt’s intra-city transportation gap using small cars as a safer, regulated alternative to tuk-tuks; single-handedly designed and built both the rider and driver apps end-to-end, from UI/UX to release.
 - Built booking and mapping with live OpenStreetMap (flutter_map) GPS tracking, debounced place autocomplete (TomTom + Nominatim), drag-pin selection with reverse geocoding, saved places (Home/Work/custom), custom pickup points, and live fare and distance estimation before booking.
 - Delivered real-time trip tracking over WebSockets: live driver location, route polyline, a full status flow (requested → driver found → arrived → in-progress → completed), trip PIN verification, driver/ETA card, cancellation with reasons, and post-trip driver rating and reviews.
 - Engineered an in-ride safety suite: hold-to-activate SOS broadcasting live location to trusted contacts via WhatsApp, a self-updating public tracking link, emergency contacts imported from the device, in-ride voice recording with upload and sharing, one-tap 122 emergency dialing, and a safety hub with guides.
 - Implemented real-time in-trip chat (text, image, and voice) with typing indicators, read receipts, client-side image compression, automatic phone-number detection to prevent off-platform contact, a mandatory driver safety-selfie gate, and auto-close on trip completion.
 - Built the driver app with document verification pending admin approval, live dispatch of incoming trip requests, a full active-trip flow (accept → navigate → PIN → complete), geofenced arrival, earnings with 7-day online-time history, an Uber Pro-style driver tier system, and a commission/balance system with auto-block and self-service recharge.
 - Handled authentication and payments: JWT with secure storage and automatic session restore, 6-digit OTP verification with resend cooldown, Egyptian mobile-number validation, password reset, and an in-app wallet with Paymob top-up (card + mobile wallet) and negative-balance booking control.
 - Developed the full backend with the help of AI tools: Node.js/Express/TypeScript REST API and WebSocket server, MongoDB with 13 Mongoose models, JWT access/refresh with bcrypt, Cloudinary uploads via multer, Nodemailer/Brevo email, HMAC-verified Paymob webhooks, and security hardening (Helmet, CORS allow-listing, rate limiting, input validation).

- Maintained a clean, feature-first Flutter codebase with Cubit state management, go_router navigation, get_it dependency injection, runtime Arabic/English switching, light/dark theming, and unit and widget tests — with zero analyzer errors or warnings.

Za2zo2a Admin Panel — Web Operations Dashboard (Founder & Sole Developer)

Flutter Web, Dart, MVVM, Cubit, Node.js, TypeScript, MongoDB, REST API

- Designed and built the complete Flutter Web owner console from UI/UX to release using an MVVM/Cubit architecture mirroring the mobile apps, with secure admin authentication, role-based access, a collapsible sidebar with live badge counts, and page-transition animations.
- Implemented full operations management: rider and driver directories with detail views, driver document verification with approve/reject reasons, blocking and unblocking, a dedicated safety-selfie moderation queue, and trip management with status tracking.
- Delivered notifications and broadcast campaigns plus a settings suite covering pricing and commission rules, service zones, support tickets, and an audit log, with uploaded media display (documents, avatars, selfies), a responsive layout, and light/dark themes.
- Built the supporting admin backend endpoints (roles, middleware, and moderation APIs) on the shared Node.js/TypeScript/MongoDB backend with the help of AI tools.

Merchandiser App — Field Merchandising App (Production, Saudi Arabia)

Flutter, Dart, GetX, REST API

- Built and shipped a production app for field merchandisers covering store visits, task management, attendance, and reporting, deployed for use in Saudi Arabia.
- Developed the full app with GetX state management and a REST API layer integrating auth, markets, branches, tasks, products, RTV, and planogram modules, with Arabic/English localization, light/dark themes, and a responsive UI.

Instagram Clone — Social Media App

Flutter, Dart, Firebase (Auth, Firestore, Storage, Messaging), Cloudinary, BLoC

- Built a full-featured social app with posts and image carousels, video reels, stories, explore/search, a follow system, likes, comments, and saved posts.
- Implemented real-time chat using Firebase (Auth, Firestore, Cloud Messaging) with Cloudinary media hosting; managed state with BLoC and Provider.

CERTIFICATIONS & COURSES

Mobile Application Development with Flutter · AMIT Learning, Nasr City

Jul 2024 – Nov 2024

- Built cross-platform mobile applications using Flutter and Dart from fundamentals to advanced topics, covering API integration, Firebase services, state management, and responsive UI design.

EDUCATION

B.Sc. Computer Science & Artificial Intelligence · Cairo Higher Institute

2022 – Jun 2026

LANGUAGES

Arabic (Native) | English (Professional Working Proficiency)